

SUPPLEMENTARY MATERIAL

A. System Conditions and Data

TABLE S1: Model Parameters Used in the Dynamic Simulation of *M. smegmatis* Growth

Parameter	Description	Value
k_C	Monod constant for glycerol [g/L]	8×10^{-2}
k_N	Monod constant for ammonia [g/L]	1×10^{-1}
k_O	Monod constant for oxygen [g/L]	1×10^{-3}
k_d	Death rate constant [1/h]	1×10^{-3}
Y_{X/CO_2}	CO ₂ yield [g/g]	0.35
Y_{X/O_2}	O ₂ yield [g/g]	1.352
$Y_{X/C}$	Biomass yield on glycerol [g/g]	0.484
$Y_{X/N}$	Biomass yield on ammonia [g/g]	21.575
X_{max}	Maximum biomass concentration [g/L]	1.4462
μ_{max}	Maximum specific growth rate [1/h]	0.15
pH_{LL}	Lower limit of pH tolerance [-]	4.6
pH_{UL}	Upper limit of pH tolerance [-]	7.4
I_{val}	pH inhibition shape parameter [-]	3
t_{lag}	Lag phase duration [h]	7

TABLE S2: Model Constants

Parameter	Chemical species or group	Value
pK_{a1}	KH ₂ PO ₄ (phosphate buffer)	6.86
pK_{a2}	Citric acid (C ₆ H ₈ O ₇)	3.13
pK_{a3}	Mono-deprotonated citric acid (C ₆ H ₇ O ₇) ⁻	4.76
pK_{a4}	Di-deprotonated citric acid (C ₆ H ₇ O ₇) ²⁻	6.40
pK_{a5}	Ammonium (NH ₃)	9.25
pK_{a6}	Glycerol (C ₃ H ₈ O ₃)	14.15
pK_{a7}	Carbon dioxide (CO ₂)	6.35
pK_{a8}	Bicarbonate (HCO ₃ ⁻)	10.33
pK_{a9}	Water (H ₂ O)	14.00
[KH ₂ PO ₄]	Phosphate buffer (monobasic) [mol/L]	2.18
[C ₆ H ₈ O ₇]	Citric acid (tricarboxylic acid) [mol/L]	2.00
pH_{alk}	Constant ions in pH charge balance [-]	7.2
$O_{2,sat}$	O ₂ saturation concentration [g/L]	7.267×10^{-3}
k_{La}	Oxygen transfer coefficient [1/h]	86.26

TABLE S3: Initial Conditions Used in the Simulation of *M. smegmatis* Growth

Variable	Description	Parameter Estimation	Validation
X_0	Biomass concentration [g/L]	0.229	0.223
C_0	Glycerol concentration [g/L]	5.389	5.922
N_0	Ammonium concentration [g/L]	0.951	1.027
$CO_{2,0}$	Carbon dioxide concentration [g/L]	4.39×10^{-4}	4.39×10^{-4}
pH_0	Dissolved oxygen concentration [g/L]	7.2	7.2
$O_{2,0}$	Dissolved oxygen concentration [g/L]	0.0001	0.0001

B. Initial Model Comparison Modified vs Extended

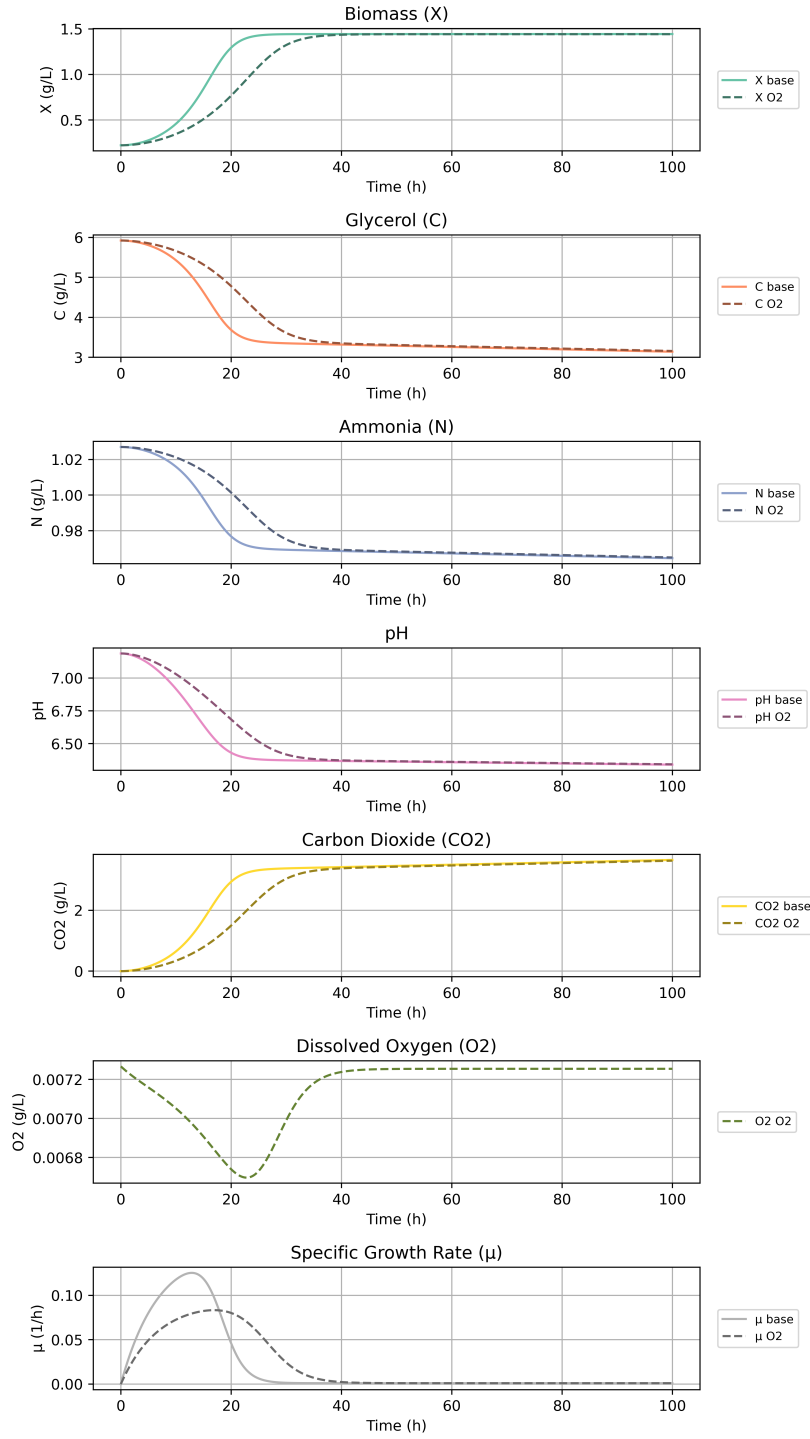


Fig. S1: Comparison between the modified model by De Witt without oxygen, and the extended model with oxygen consumption on hypoxic conditions.

C. Base Model Analysis

Model 1: Fitting ['YX_CO2', 'YX_C', 'Xmax', 'pH_LL'] vs. Validation Data

k_C : 0.0800
 k_N : 0.1000
 k_O : 0.0010
 k_d : 0.0010
 YX_{CO2} : 0.2702
 YX_{O2} : 1.3520
 YX_C : 0.5490
 YX_N : 21.5750
 X_{max} : 1.3071
 μ_{max} : 0.1500
 pH_{LL} : 2.0000
 pH_{UL} : 7.4000
 I_{val} : 3.0000
 t_{lag} : 7.0000

AIC = -135.3691

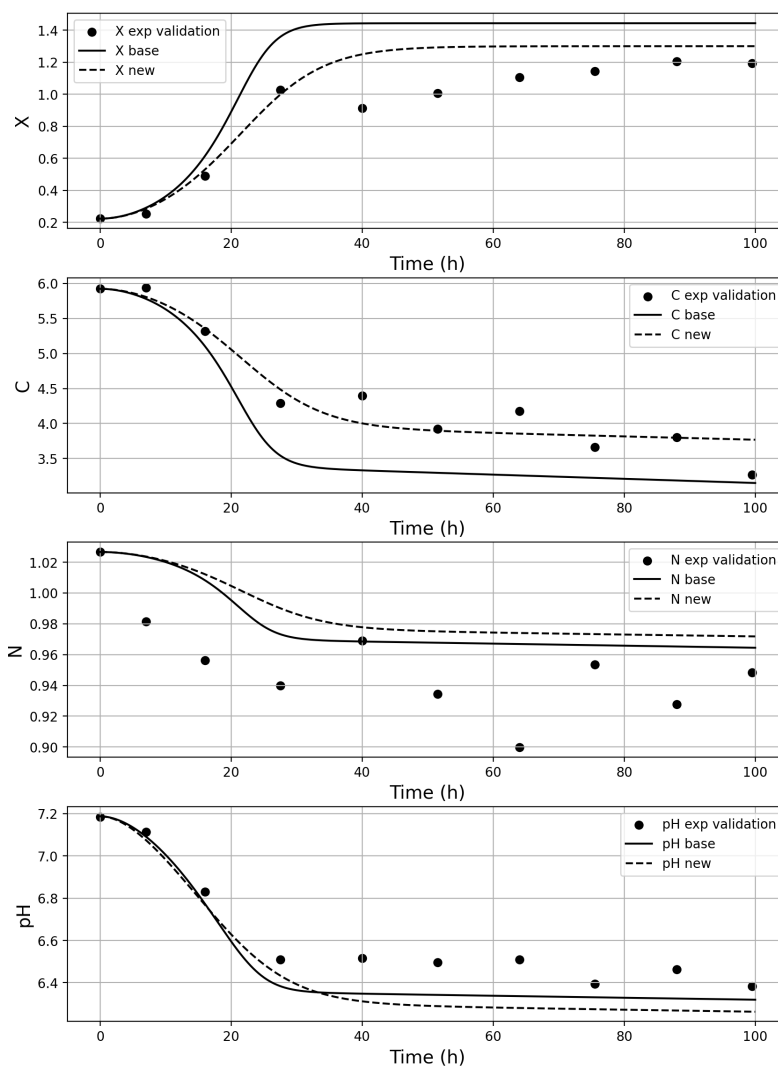


Fig. S2: Validation Experiment Points vs De Witt's Model

Residuals Analysis Model 1

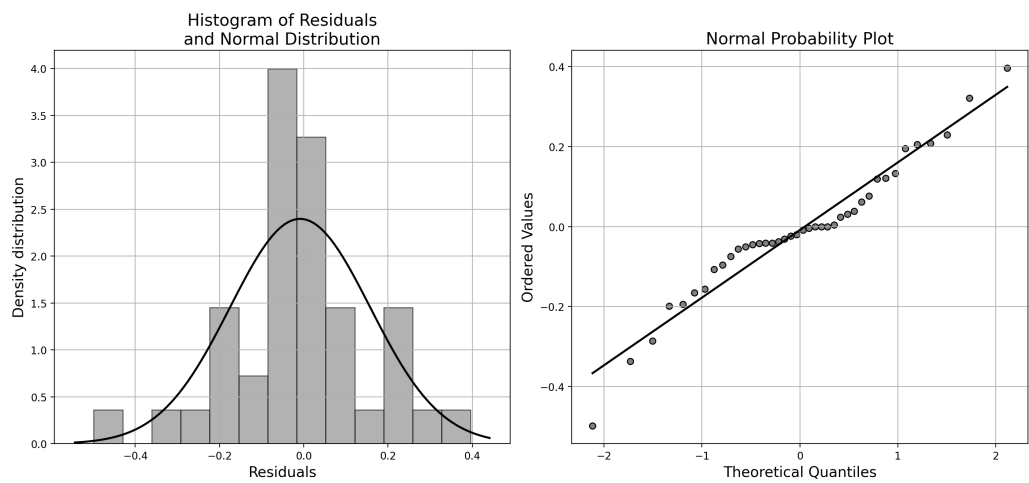


Fig. S3: Normal Distribution Fitting and Normal Probability Plot for De Witt's Model

Parameter Sensitivities Overview Model 1

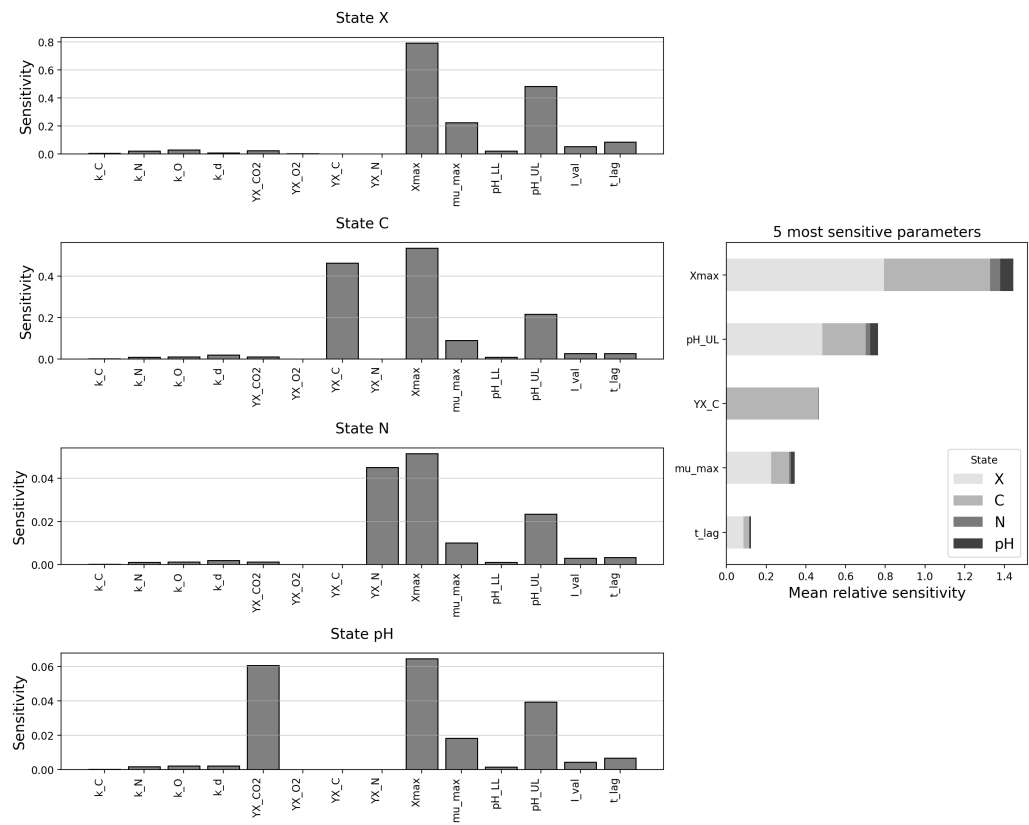


Fig. S4: Sensitivity Analysis of Parameters for De Witt's Model

D. Base Model Analysis

k_C: 0.0800
k_N: 0.1000
k_O: 0.0010
k_d: 0.0010
YX_CO2: 0.3500
YX_O2: 1.3520
YX_C: 0.4840
YX_N: 21.5750
Xmax: 1.4462
mu_max: 0.1500
pH_LL: 4.6000
pH_UL: 7.4000
l_val: 3.0000
t_lag: 7.0000

AIC = nan

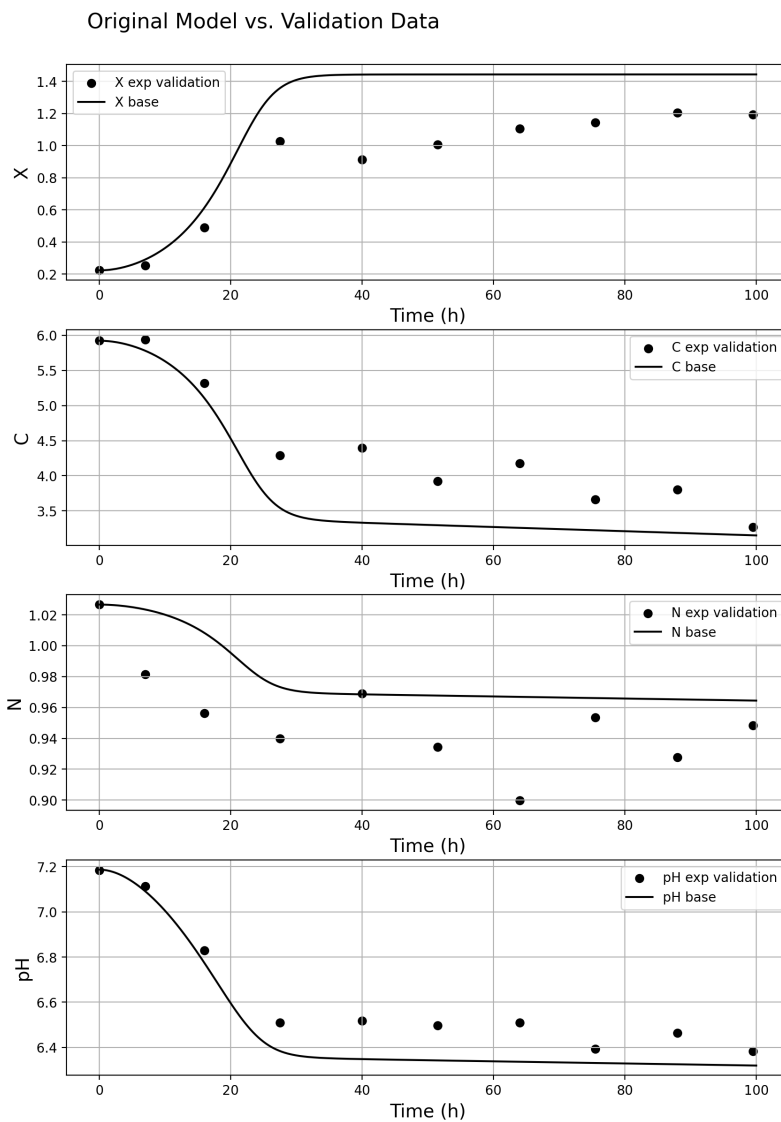


Fig. S5: Validation Experiment Points vs Base Model

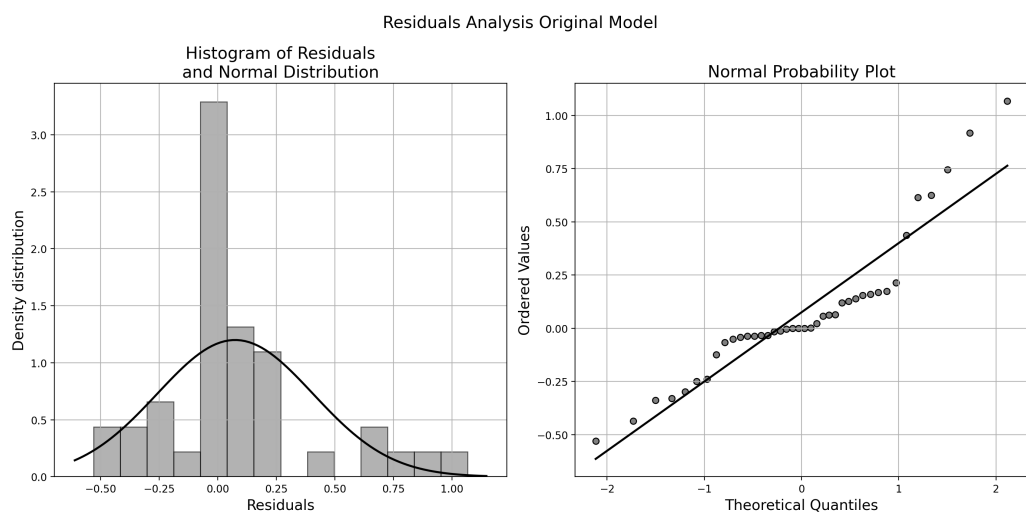


Fig. S6: Normal Distribution Fitting and Normal Probability Plot for Base Model

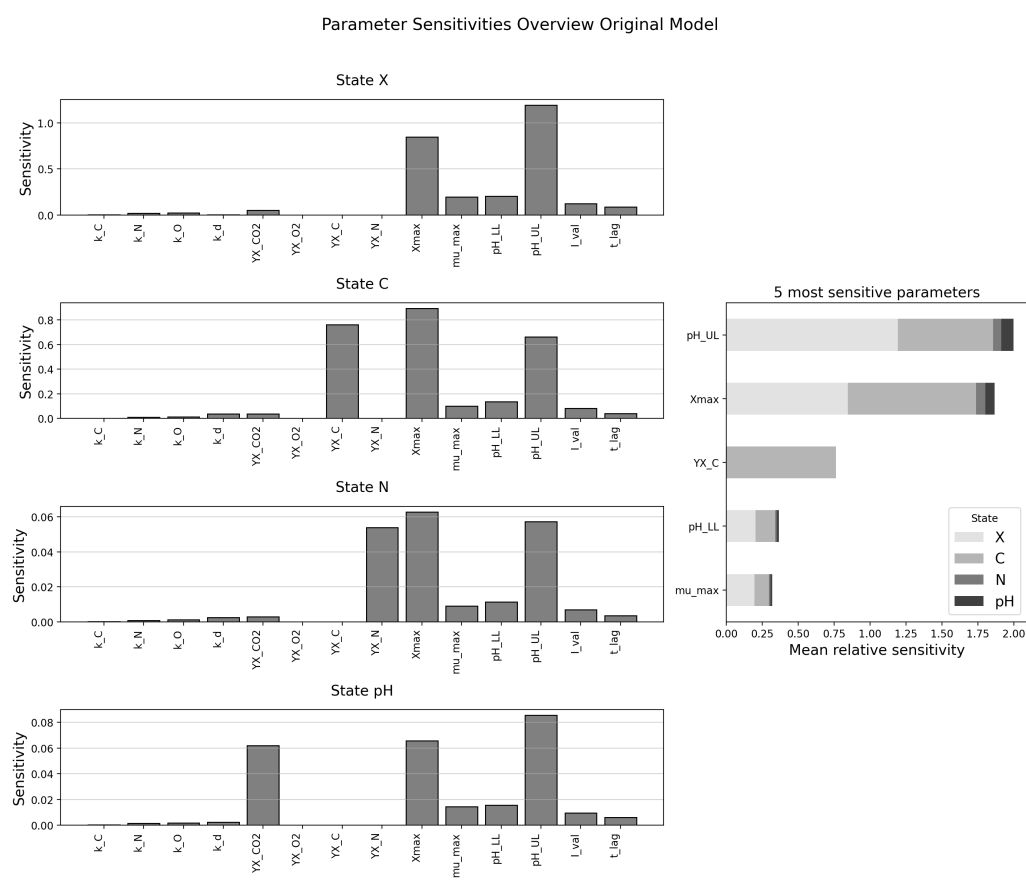


Fig. S7: Sensitivity Analysis of Parameters for Base Model

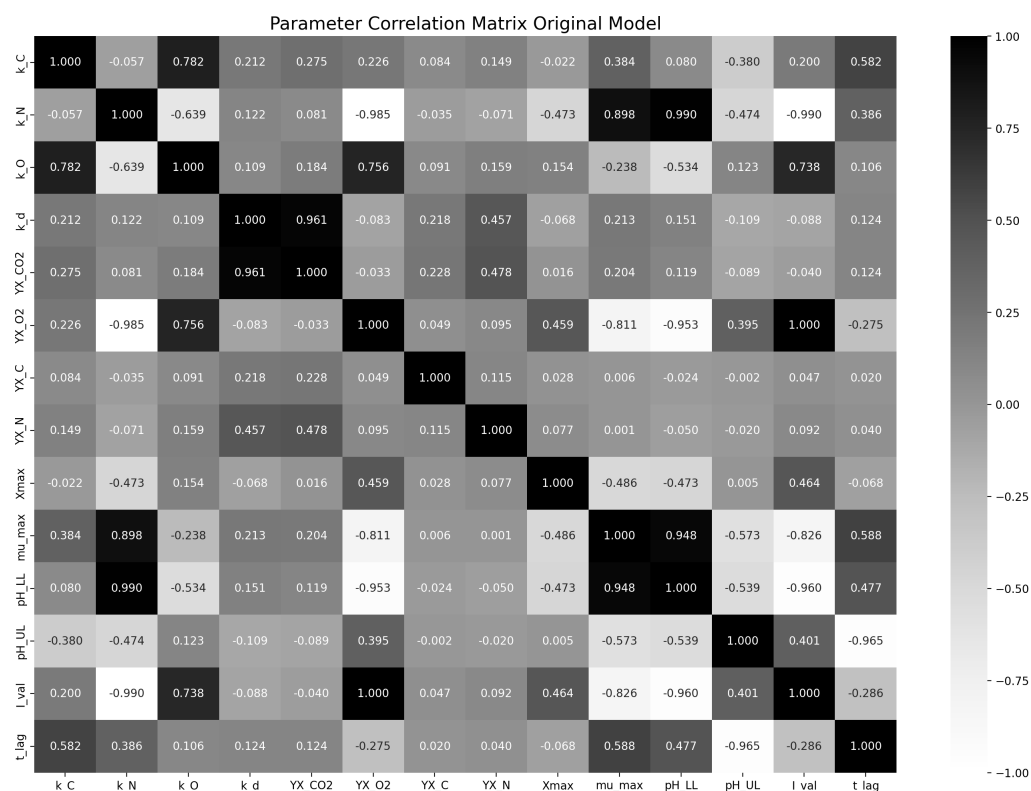


Fig. S8: Correlation Matrix for Base Model

E. Exploring Feasible Boundaries

F. Parameter Estimation Results

1) *First Iteration:* Use of the DAE package with calibration of $Y_{X/C}$, $Y_{X/N}$, X_{max} , μ_{max} , pH_{UL} was performed with the values seen on Figure S9. 50 iterations were performed for all proposed models allowing many solutions to come up as possible best fits.

```
1 from RUN_functions import RUN_ALL_COMBOS
2
3 parameters = {
4     # 'k_C': [0.01, 0.1],
5     # 'k_N': [0.0105, 0.5],
6     # 'k_O' : [0.0001, 0.009],
7     # 'k_d': [0.001, 0.005],
8     # 'YX_CO2': [0.1, 1.0],
9     # 'YX_O2' : [0.5, 5],
10    # 'YX_C': [0.35, 0.8],
11    'YX_N': [10.4, 10.6],
12    'Xmax': [1.15, 1.17],
13    'mu_max': [0.13, 0.15],
14    # 'pH_LL': [1, 4.8],
15    'pH_UL': [5.5, 21],
16    # 'I_val': [1, 5],
17    # 't_lag': [6, 10]
18 }
19
20 df_all = RUN_ALL_COMBOS(
21     iterations=50,
22     initial_given_data=parameters,
23     folder='Calibration_Combos',
24     file_name='Calibration_Combos.csv',
25     updated_parameters=None,
26     min_k = 3,
27     max_k = 3,
28 )
29
```

Fig. S9: Code for first calibration

The best results were obtained by estimating only X_{max} (1.152807872) and μ_{max} (0.132614019), looking at an AIC value of -155.6284133 as seen on Figure S10. The overall NMRSE is 19.534 %.

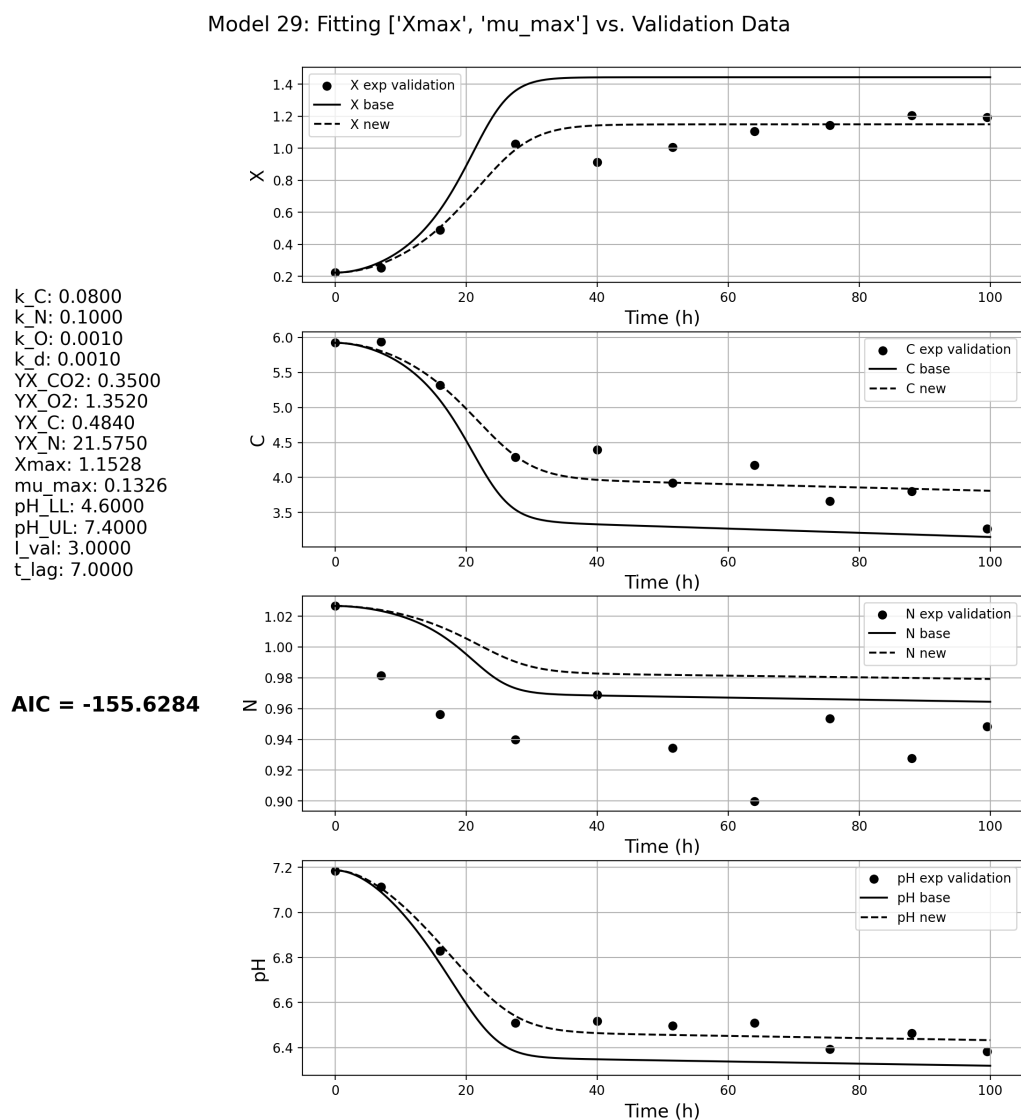


Fig. S10: Simulation Results for Model 29 on First Calibration

It can be seen that in Amonnia dynamics, the experimental values are not well simulated. This is due to the term $Y_{X/N}$ being necessary to increase or decrease where the stationary phase will begin on its consumption.

The next best model considering now $Y_{X/N}$ as a must calibrate parameter is when estimating X_{max} (1.176285581), pH_{UL} (6.820530661) and $Y_{X/N}$ (12.08287455), with an AIC of -155.2800713 as seen on S11. The overall NMRSE is 19.135 %.

Model 42: Fitting ['YX_N', 'Xmax', 'pH_UL'] vs. Validation Data

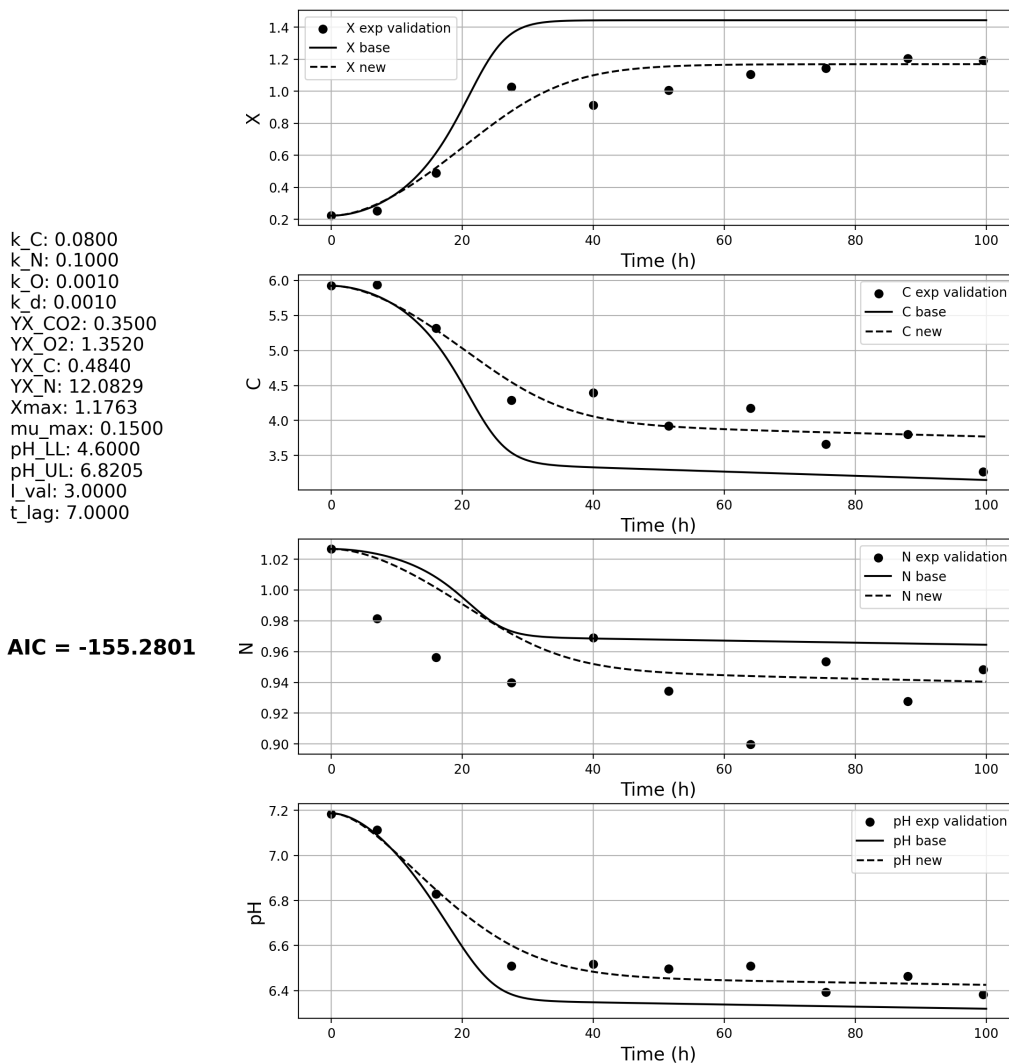


Fig. S11: Simulation Results of Model 42 $Y_{X/N}$, X_{max} and pH_{UL} on First Calibration

Although validation metrics support this model very well, a visual inspection allows to see a flawed representation of how bacterial dynamics are. The slopes of growth and consumption are smaller than what the validation data represent.

One combination that allows for a visually acceptable model is the calibration of $Y_{X/N}$ (14.98048135), X_{max} (1.162267901) and μ_{max} (0.126263825). The overall AIC value is -153.7399085 as seen on S12.

Model 10: Fitting ['YX_N', 'Xmax', 'mu_max'] vs. Validation Data

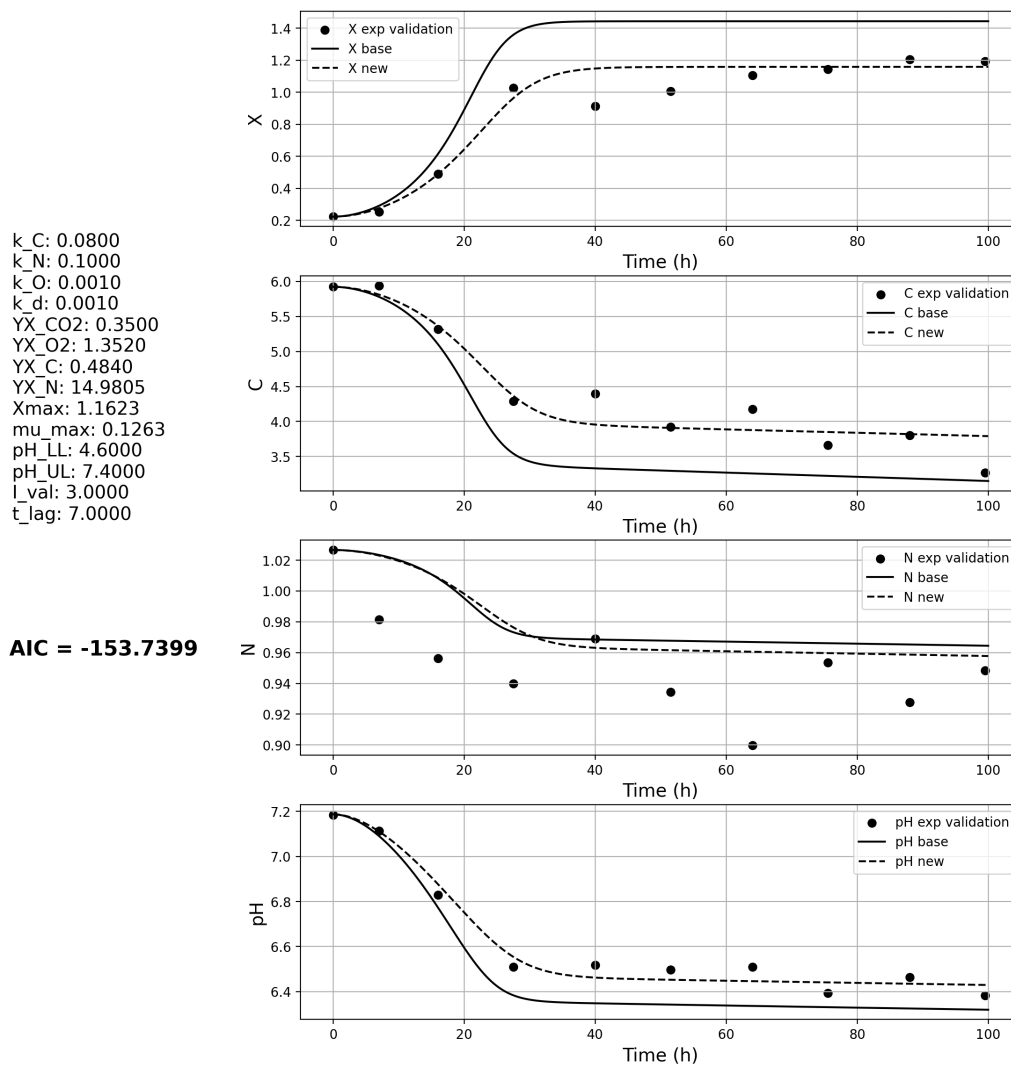


Fig. S12: Simulation Results of Model 10 Fixing $Y_{X/N}$, X_{max} and μ_{max} on First Calibration

This simulation represents better the dynamics.

2) *Second Iteration*: Considering that $Y_{X/N}$, X_{max} and μ_{max} are the parameters to estimate, due to X_{max} being the limit of how much growth of biomass can be, $Y_{X/N}$ the parameter that sets where the stationary phase of N consumption begins, and μ_{max} the value to better fit the growth speed. pH_{UL} limited the dynamics, therefore it was better left fixed.

While iterating with narrower limits, the code on Figure S13 was performed.

```

1 from RUN_functions import RUN_ALL_COMBOS
2
3 parameters = {
4     # 'k_C': [0.01, 0.1],
5     # 'k_N': [0.0105, 0.5],
6     # 'k_O' : [0.0001, 0.009],
7     # 'k_d': [0.001, 0.005],
8     # 'YX_CO2': [0.1, 1.0],
9     # 'YX_O2' : [0.5, 5],
10    # 'YX_C': [0.35, 0.8],
11    'YX_N': [10.4, 10.6],
12    'Xmax': [1.15, 1.17],
13    'mu_max': [0.13, 0.15],
14    # 'pH_LL': [1, 4.8],
15    # 'pH_UL': [5.5, 21],
16    # 'I_val': [1, 5],
17    # 't_lag': [6, 10]
18 }
19
20 df_all = RUN_ALL_COMBOS(
21     iterations=50,
22     initial_given_data=parameters,
23     folder='Calibration_Combos_Results',
24     file_name='Calibration_Combos_Results.csv',
25     updated_parameters=None,
26     min_k = 3,
27     max_k = 3,
28 )
29

```

Fig. S13: Code for second calibration

Here the final result is obtained in between many similar models. The best one was model 69 as seen on Figure S14.

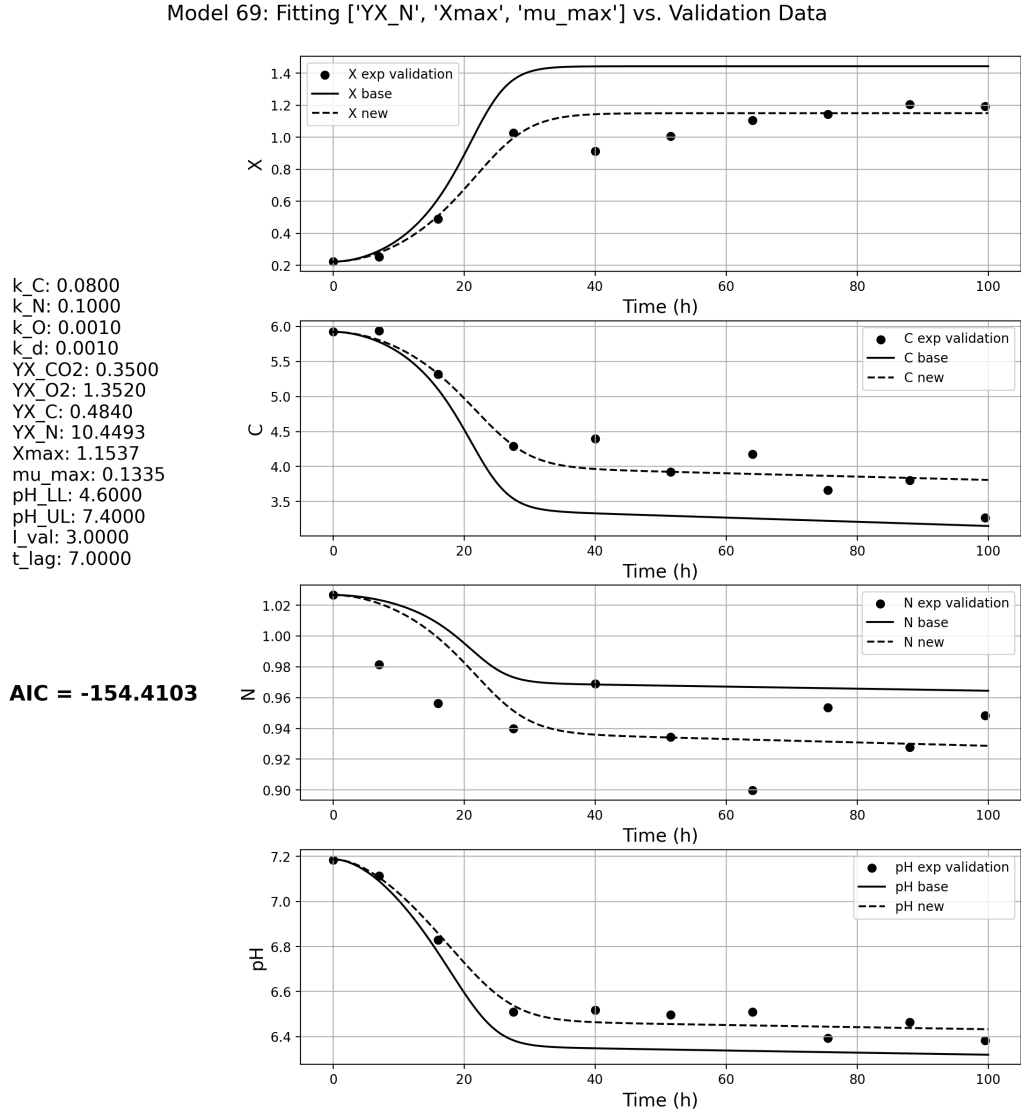


Fig. S14: Simulation Results of Model 69 Fixing $Y_{X/N}$, X_{max} and μ_{max} on Second Calibration

With values $Y_{X/N} = 10.4493286300$, $X_{max} = 1.1536938460$ and $\mu_{max} = 0.1334781790$ the best numerically and visual model is chosen. Its AIC of -154.4103054560.

Correlation seen on Figure S15 indicates three independent parameters— $Y_{X/C}$, $Y_{X/N}$, and X_{max} —and three distinct groups of highly correlated parameters:

- $\{k_C, k_N, k_O, Y_{X/O_2}, pH_{LL}, I_{val}, \mu_{max}\}$,
- $\{pH_{UL}, t_{lag}\}$,
- $\{k_d, Y_{X/CO_2}\}$.

Here, the estimated parameters are clearly independent on a great length from each other, validating results. Also, interestingly, it is possible to observe how μ_{max} is responsible for great part of the model, specially for oxygen dynamics.

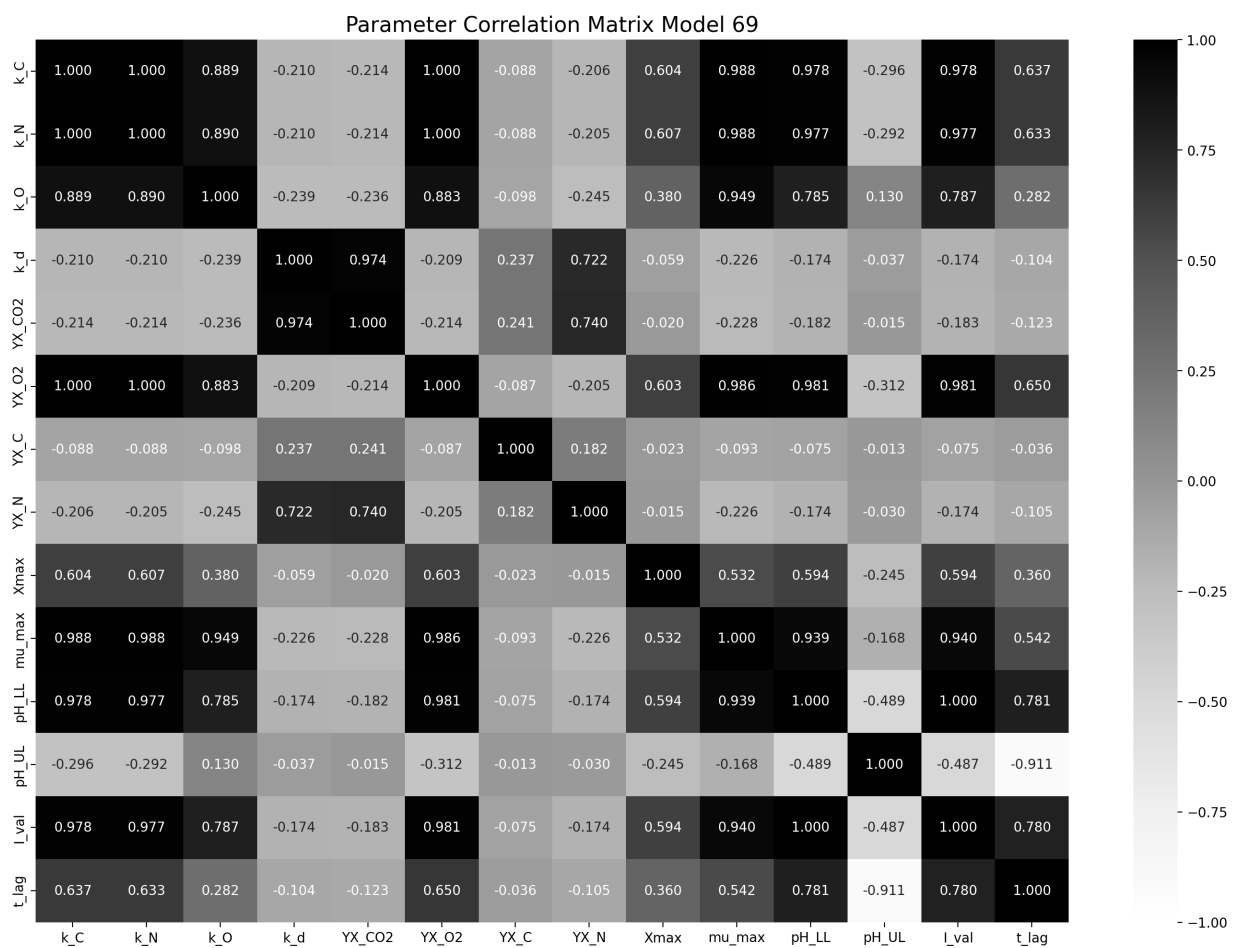


Fig. S15: Correlation Matrix for Best Simulation

Parameter Sensitivities Overview Model 69

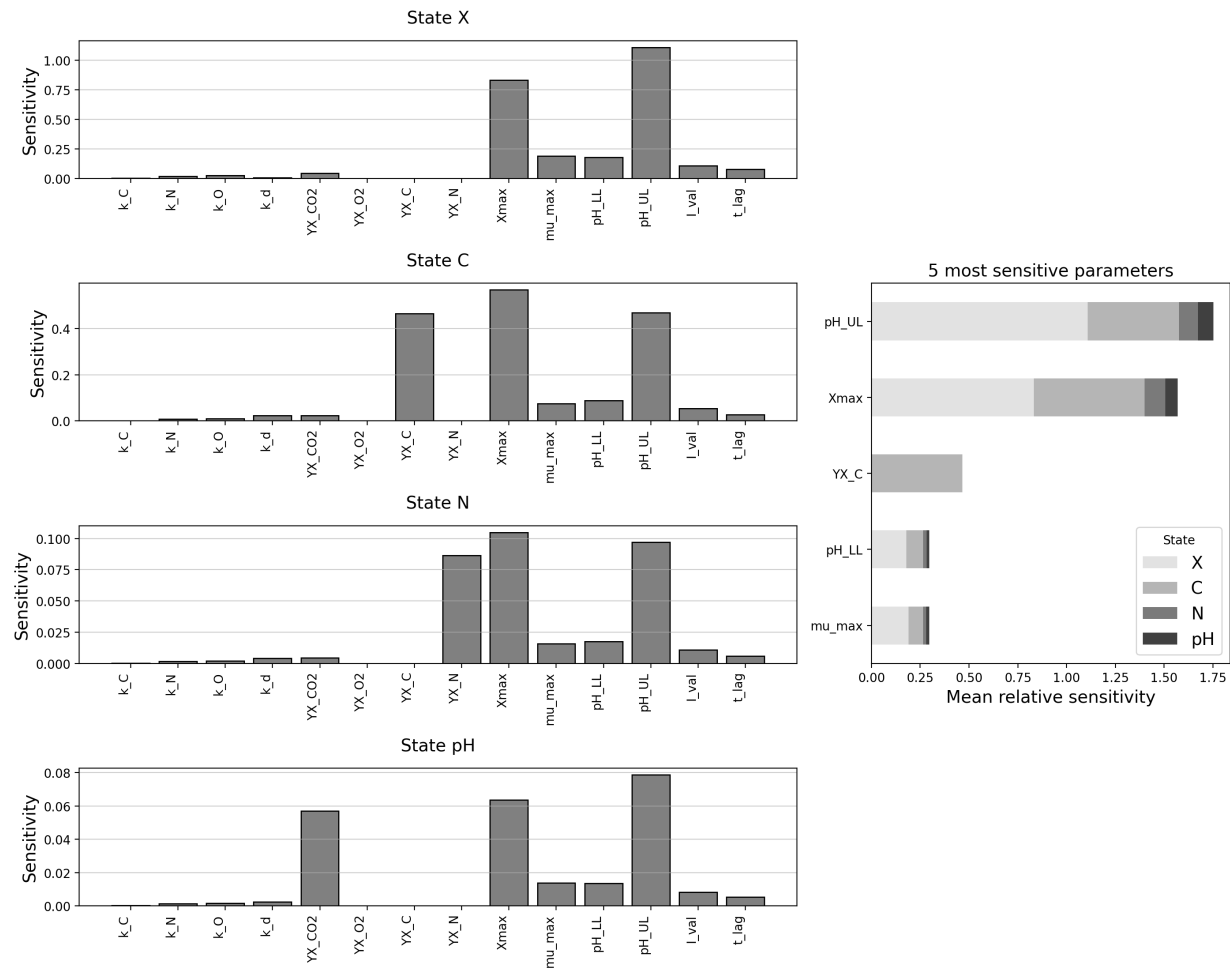


Fig. S16: Sensitivity for Best Simulation

Residuals Analysis Model 69

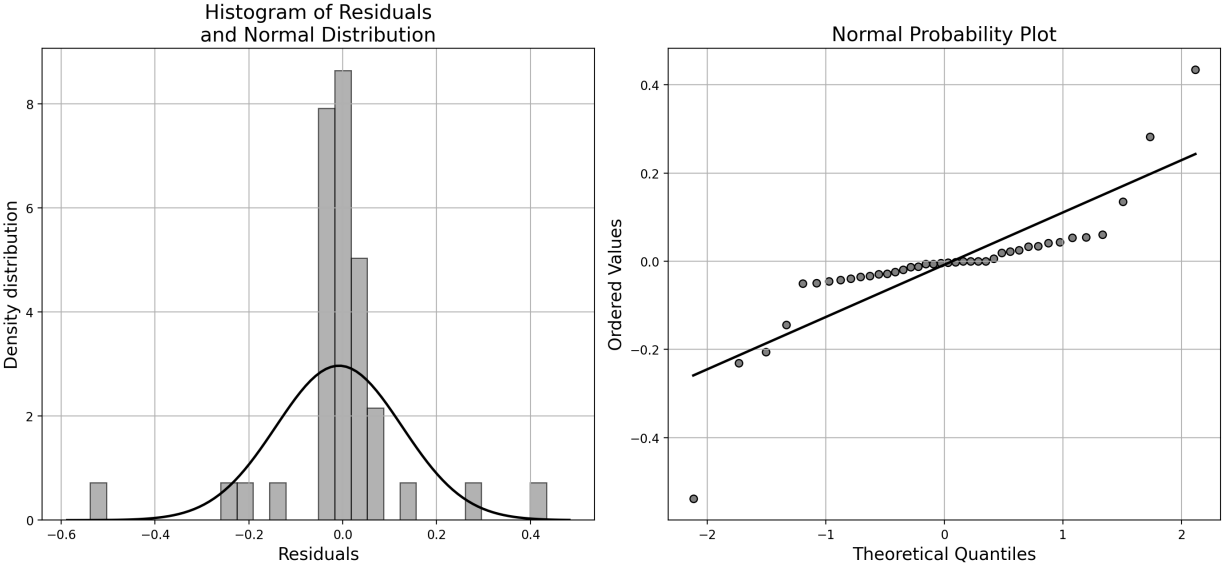


Fig. S17: Residuals for Best Simulation